

1. A railway truck S of mass 20 tonnes is moving along a straight horizontal track when it collides with another railway truck T of mass 30 tonnes which is at rest. Immediately before the collision the speed of S is 4 m s^{-1}
As a result of the collision, the two railway trucks join together.

Find

- (a) the common speed of the railway trucks immediately after the collision, (2)
- (b) the magnitude of the impulse exerted on S in the collision, stating the units of your answer. (3)
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2.

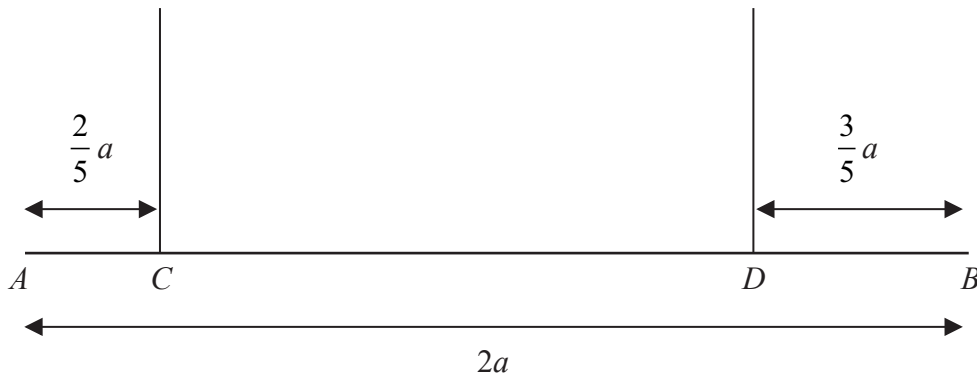


Figure 1

A uniform rod AB has length $2a$ and mass M . The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D ,

where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

A particle P is placed on the rod at B .

The rod remains horizontal and in equilibrium.

(a) Find, in terms of M , the largest possible mass of the particle P

(3)

Given that the mass of P is $\frac{1}{2}M$

(b) find, in terms of M and g , the tension in the string that is attached to the rod at C .

(3)

3.

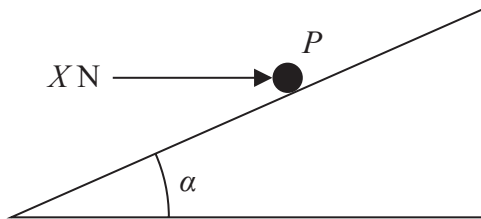


Figure 2

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

(a) Show that when $X = 14.7$ there is no frictional force acting on P

(3)

The coefficient of friction between P and the plane is 0.5

(b) Find the smallest possible value of X .

(8)

4.

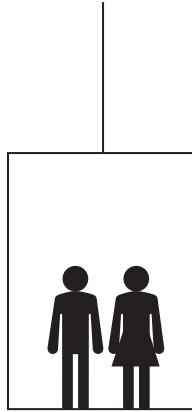


Figure 3

Two children, Alan and Bhavana, are standing on the horizontal floor of a lift, as shown in Figure 3.

The lift has mass 250 kg . The lift is raised vertically upwards with constant acceleration by a vertical cable which is attached to the top of the lift. The cable is modelled as being light and inextensible. While the lift is accelerating upwards, the tension in the cable is 3616 N .

As the lift accelerates upwards, the floor of the lift exerts a force of magnitude 565 N on Alan and a force of magnitude 226 N on Bhavana.

Air resistance is modelled as being negligible and Alan and Bhavana are modelled as particles.

(a) By considering the forces acting on the lift only, find the acceleration of the lift.

(3)

(b) Find the mass of Alan.

(3)

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (a) Find the distance travelled by the ball while its speed is less than 14.7 m s^{-1} (3)
- (b) Find the time for which the ball is moving with a speed of more than 29.4 m s^{-1} (3)
- (c) Sketch a speed-time graph for the motion of the ball from the instant when it is projected from A to the instant when it hits the ground. Show clearly where your graph meets the axes. (3)
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6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j}) \text{ N}$ and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j}) \text{ N}$, where p and q are constants, are applied to A .

Given that A moves in the direction of the vector $(\mathbf{i} - 2\mathbf{j})$,

(a) show that $2p + q - 4 = 0$

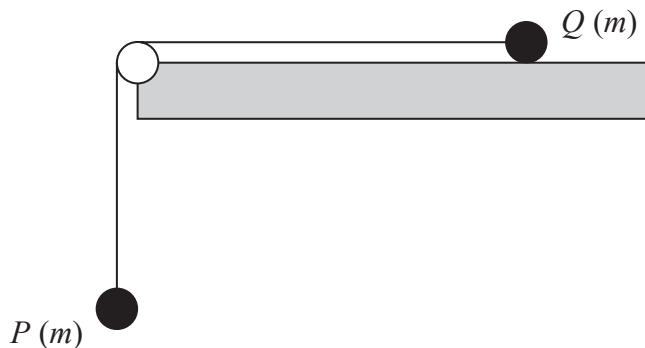
(4)

Given that $p = 5$

(b) find the speed of A at time $t = 4$ seconds.

(5)

7.

**Figure 4**

A particle P of mass m is attached to one end of a light inextensible string. Another particle Q , also of mass m , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle Q is held at rest on the table and particle P hangs vertically below the pulley with the string taut, as shown in Figure 4.

The pulley, P and Q all lie in the same vertical plane.

The coefficient of friction between Q and the table is μ , where $\mu < 1$

Particle Q is released from rest.

The tension in the string before Q hits the pulley is kmg , where k is a constant.

(a) Find k in terms of μ .

(7)

Given that Q is initially a distance d from the pulley,

(b) find, in terms of d , g and μ , the time taken by Q , after release, to reach the pulley.

(4)

(c) Describe what would happen if $\mu \geq 1$, giving a reason for your answer.

(2)

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two ships, A and B , are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$ and the velocity of B is $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of A .

(2)

The ships are modelled as particles.

At 12 noon, A is at the point with position vector $(-9\mathbf{i} + 6\mathbf{j})\text{ km}$ and B is at the point with position vector $(16\mathbf{i} + 6\mathbf{j})\text{ km}$.

At time t hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{ km}$$

- (b) Find the value of p and the value of q .

(7)

- (c) Find the bearing of A from B when the ships are 15 km apart, giving your answer to the nearest degree.

(7)
